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Author(s): Jason T. Bried and Gary N. Ervin

Source: Southeastern Naturalist, 4(4):731-744. 2005.

Published By: Eagle Hill Institute

DOI: [http://](http://dx.doi.org/10.1656/1528-7092(2005)004[0731:DOAOAL]2.0.CO;2)

[dx.doi.org/10.1656/1528-7092\(2005\)004\[0731:DOAOAL\]2.0.CO;2](http://dx.doi.org/10.1656/1528-7092(2005)004[0731:DOAOAL]2.0.CO;2)

URL: [http://www.bioone.org/doi/](http://www.bioone.org/doi/full/10.1656/1528-7092%282005%29004%5B0731%3ADOAOAL%5D2.0.CO%3B2)

[full/10.1656/1528-7092%282005%29004%5B0731%3ADOAOAL%5D2.0.CO%3B2](http://www.bioone.org/doi/full/10.1656/1528-7092%282005%29004%5B0731%3ADOAOAL%5D2.0.CO%3B2)

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## Distribution of Adult Odonata Among Localized Wetlands in East-central Mississippi

JASON T. BRIED<sup>1,2,\*</sup> AND GARY N. ERVIN<sup>1</sup>

**Abstract** - We measured species richness and composition of adult Odonata and inferred habitat preferences among man-made wetland sites and surrounding tracts of natural bottomland forest. Cumulative species richness and composition were described by proportion coefficients and beta diversity indices. The three man-made sites provided open space resources, and more species were observed in each than in the floodplain forest. Twenty-nine of 42 species documented over a four-month period were observed in only one or two of the four wetlands studied. Large differences in species assemblages between the immediately adjacent ditch and marsh sites were the best evidence for high habitat affinity because distance and structural barriers to movement were absent. Such compositional asymmetry may reflect differential vegetative and reproductive suitability of the habitats. Results suggest that the open-canopy wetlands supported higher diversity of adult Odonata, and that distinct odonate assemblages were found among different habitat types in this floodplain wetland complex.

### Introduction

Forested riparian ecosystems in the southeastern United States often include lake impoundments, greentree reservoirs, and moist-soil impoundments (Smith et al. 1989) among a natural mosaic of streams, sloughs, flats, low depressions, and oxbow lakes (Mitsch and Gosselink 2000). Such systems of localized (high density) habitats or sub-ecosystems may be adjacent to each other and are thus ideal settings for studies of habitat distribution and preference.

Relatively little is known about habitat distributions in adult, versus larval, stages of wetland insects (King and Wrubleski 1998, Wissinger 1999), although such information would contribute significantly to improvements in biological assessment and monitoring methodologies. Easily observed species with high fidelity to a narrow range of environmental conditions could serve as useful rapid assessment indicators of ecological integrity. Knowledge of occurrence patterns in adult odonate (damselfly and dragonfly) assemblages may be useful for wetland assessment.

Adult Odonata link terrestrial and aquatic food webs and function both as opportunistic predators and food items for wetland vertebrates (e.g., Kennedy 1950, King and Wrubleski 1998, Orians and Horn 1969, Turner and McCarty 1998, Wissinger 1988a). Odonata have shown potential as

<sup>1</sup>Department of Biological Sciences, Mississippi State University, PO Box GY, Mississippi State, MS 39762. <sup>2</sup>Current address - The Nature Conservancy, Eastern New York Conservation Office, 200 Broadway, Suite 301, Troy, NY 12180. \*Corresponding author - jtbried@tnc.org.

species richness indicators for overall site invertebrate (Briers and Biggs 2003), predatory invertebrate (Davis et al. 1987), and macrophytic (Sahlen and Ekestubbe 2001) richness. They also have served as conservation indicators for assessing the ecological integrity of aquatic ecosystems (e.g., Burton et al. 1999, Chovanec and Raab 1997, Chovanec and Waringer 2001, Clausnitzer 2003, Stewart and Samways 1998). Odonata are ecologically sensitive to physical alterations of habitat structure (Jonsen and Taylor 2000a,b), such as upland forest management (Kinvig and Samways 2000, Rith-Najarian 1998, Sahlen 1999), road development (King et al. 2000, Riffell 1999), and trampling of shoreline vegetation (Muller et al. 2003, Wood et al. 2001). They have long life cycles relative to other insects, taxonomy is well defined, and natural history is described for many species (e.g., Wissinger 1988b). Current known geographic distributions are mapped for all North American species (Donnelly 2003, 2004a,b). Furthermore, adult stages are easily observed and aesthetically pleasing so as to attract public conservation interest (Corbet 1993, Samways 1993) and indirectly facilitate preservation of wetland habitat and therefore other wetland-dependant biota (i.e., odonates as “umbrellas”).

We sampled a floodplain odonate assemblage in the southeastern United States in order to describe localized distribution and habitat preferences. Species richness and composition were compared among an herbaceous seasonal ditch, managed moist-soil marsh, shallow fringe of an adjacent reservoir, and tracts of surrounding bottomland forest. We compared communities in the ditch versus the adjacent moist-soil wetlands to test for habitat partitioning at midday during peak flight season (May–August).

## Methods

### Study site

The study site was in the Noxubee National Wildlife Refuge (NWR), located in east-central Mississippi. Noxubee NWR includes over 5300 ha of forested riparian wetlands, including 275 ha of managed bottomlands known as greentree reservoirs (Duffy and LaBar 1994). The site included moist-soil impoundments, a seasonally flooded ditch dominated by herbaceous vegetation, a lake littoral fringe, and bottomland hardwood forests. All wetlands were within 4.7 km<sup>2</sup> of the floodplain of Jones Creek, a first-order tributary of the Noxubee River to the north.

Moist-soil impoundments comprised an 80-ha complex of 13 marsh units partitioned by earthen berms. These units have been managed since construction in 1983 to promote habitat for overwintering and migratory waterfowl (Gray et al. 1999). Refuge personnel typically drain the impoundments in late spring and flood them to 0.5–1.0 m in mid-fall. Planting, mowing, and disking are used to expose seed banks, increase primary productivity, and enhance food supplies for waterfowl. During 2003, when these surveys were conducted, drainage was initiated 2 June (L. Williamson, Refuge Manager, pers. comm.). Dominant macrophytes of the moist-soil

units included *Brunnichia ovata* (Walt.) Shinnery, *Carex lupulina* Muhl. ex Schk. in Willd., *Cephalanthus occidentalis* L., *Panicum rigidulum* Nees, *Polygonum* spp., and *Salix nigra* Marsh.

A ditch ( $\approx 6$  m width) extends around the moist-soil management area. Hydrologic inputs include direct precipitation and seasonal flow from the surrounding floodplain forest. Prior to drawdown on 8 May, 13 May, and 28 May, mean depth was 4.6, 4.3, and 4.2 cm, respectively, (max 12 cm) as measured from 20 fixed points. The ditch plant community includes a mixture of grasses (e.g., *Panicum scoparium* Lam., *Paspalum urvillei* Steud.), sedges (*Carex* spp., *Cyperus* spp., *Rhynchospora* spp.), rushes (*Juncus* spp.), forbs (e.g., *Chamaecrista fasciculata* (Michaux) Greene, *Helianthus angustifolius* L., *Solidago altissima* L.), and vines (e.g., *Campsis radicans* (L.) Seem., *Brunnichia ovata* (Walt.) Shinnery).

Floodplain forest borders the management area on three sides and a fire-managed tract of *Pinus taeda* L. (loblolly pine) abuts the southern edge of the property. The floodplain forest is composed of characteristic southeastern bottomland hardwoods including *Acer rubrum* L. (red maple), *Liquidambar styraciflua* L. (sweet gum), *Carpinus caroliniana* Walt. (hornbeam), *Nyssa* spp. (tupelos), *Quercus* spp. (oaks), and *Carya* spp. (hickories). Inundation is continuous in deep-water channels, oxbows, and cypress-tupelo sloughs, but sporadic in the remaining forest throughout the year. Ross Branch Reservoir is a 17 ha lake impoundment situated near the southwest corner of the complex. Maximum depth is approximately 3 m and the shallow fringes support emergent and floating-leaved plants, such as *Eleocharis quadrangulata* (Michx.) Roemer & J.A. Shultes and *Nymphaea odorata* Ait.

### Odonata census

Odonata were recorded from May–September 2003 in Ross Branch Reservoir, the moist-soil units, bottomland hardwood forest, and ditch habitat. Table 1 provides general sampling designs for each habitat.

Surveys of all sites were conducted on foot by the same observer (J.T. Bried) throughout the study period. For transect surveys (Table 1), species flying or perched within  $\approx 10$  m of the transect route were documented. The northern margin of Ross Branch Reservoir was patrolled 400 m with doubling-back.

In the bottomland forest, species were recorded along three parallel 400–1000 m transects, each 200 m apart, in three floodplain tracts north of the management complex. Each tract was surveyed on a separate date.

Table 1. Description of sampling effort in each of four wetland sites surveyed for adult Odonata in east-central Mississippi. NA = not applicable.

| Sites                 | Plot area                  | Transect length | Sampling frequency     |
|-----------------------|----------------------------|-----------------|------------------------|
| Bottomland forest     | $\approx 0.3 \text{ km}^2$ | 2.25 km         | 8 dates (May–Aug)      |
| Moist-soil marsh      | $\approx 0.3 \text{ km}^2$ | NA              | 17 dates (5 May–5 Sep) |
| Perimeter ditch       | $\approx 1.2 \text{ km}^2$ | NA              | 17 dates (8 May–5 Sep) |
| Ross Branch Reservoir | NA                         | 0.8 km          | 9 dates (May–Aug)      |

Additional surveys occurred on five dates within a forested plot ( $\approx 0.3 \text{ km}^2$ , Table 1) near the moist-soil complex. All bottomland surveys lasted 2–5 hrs (during 10:00–15:00 hr) on each of the eight total visits.

Three moist-soil units and the perimeter ditch were monitored on 13 May, 28 May, 8 June, 13 June, 18 June, 2 July, 12 July, 24 July, 2 August, 13 August, 23 August, and 5 September 2003, with five additional dates ( $\pm 3 \text{ d}$  apart) each. A pair of 2- x 24-m plots, delimited by red stake-wire flagging, was constructed within each moist-soil unit. Plots were at least 40 m apart and were oriented adjacent and parallel to the levee. All odonates observed to fly over or perch inside the plot were recorded during a one-way walk along the plot edge. The ditch census was conducted along a 200-m linear segment. All odonates flying across the  $\approx 6 \text{ m}$  width or along the length, or perched within, were recorded. Species abundance of dragonflies (not damselflies) was estimated in the plot areas of both sites. Sites were sampled during 10:00–12:45 hr (10–15 min. per moist-soil plot followed by 30 min. in the ditch) on each date. The fixed survey area was approximately four times greater in the ditch ( $\approx 1200 \text{ m}^2$ ) than in the moist-soil units ( $288 \text{ m}^2$ ), and moist-soil units together were sampled for 2–3 times longer duration.

Moist-soil and ditch habitats were sampled more frequently than the reservoir or bottomlands. However, extensive areal coverage in the bottomlands and reservoir, and qualitative data collections of a conspicuous and long-lived insect group (therefore, fewer observations required to detect overall presence), helped validate the comparisons. We acknowledge possible sampling bias, but assume the observed diversity patterns reflect real differences and are not artifact of differences among sampling regimes.

Surveys were conducted on warm, sunny days. Average air temperature ( $^{\circ}\text{C}$ ) and relative humidity were taken during every survey in the ditch and moist-soil sites. Maximum and average wind speed (m/s) was recorded for 2 min at the end of each interval (Kestrel 4000 Pocket Weather Station). Cloud cover was estimated as sunny ( $< 5\%$  of survey time occurred under cloud cover), mostly sunny (5–30% cover), partly cloudy (30–60% cover), mostly cloudy (60–95% cover), or overcast ( $> 95\%$  cover) during each sampling visit. Weather summaries are based on data from the perimeter ditch and moist-soil units since these sites were surveyed most frequently and encompass the sampling periods at the other two sites.

Periodic site visits between census dates helped identify questionable individuals, which were usually cryptic females or teneral (newly emerged adults) of damselflies. During these checks, questionable damselflies were captured and identified in-hand or via laboratory examination using Westfall and May (1996). Vouchers obtained for most species were deposited in the Mississippi Entomological Museum, Mississippi State University.

### **Proportion coefficients and $\beta$ -diversity**

Presence-absence of species was compared among the four types of wetlands. Sørensen's ( $C_s$ ) index compares species similarity between two

sites based on cumulative species lists and degree of species overlap (Magurran 1988). Sørensen's index is scored from 0 to 1, where values closer to 0 indicate greater species similarity between sites. Colwell and Coddington (1994) argued that dissimilarity provides a better description of inter-site species relationships. Their dissimilarity measure is essentially a reformulation of the Jaccard model, such that greater ecological distance corresponds to lower values within the 0 to 1 output range:

$$C_{jk} = (S_j + S_k - 2V_{jk}) / (S_j + S_k - V_{jk}) \quad \text{or,}$$

$$C_{jk} = 1 - C_j$$

where,  $S_j$  is richness of one site and  $S_k$  is richness of another,  $V_{jk}$  is the number of species shared between the two site lists, and where  $C_j$  is Jaccard's index. Dissimilarity indices have gained favor in ecology (e.g., Howard et al. 1998) and we provide results from both  $C_s$  similarity and  $C_{jk}$  dissimilarity.

Beta diversity ( $\beta$ ) is defined as species composition along an environmental gradient ("community turnover") or as comparative assemblage patterns among sites (Primack 1998, Whittaker 1972). High beta diversity corresponds to high dissimilarity among sample units. We calculated two indices to indicate community heterogeneity among sample units (the four wetland types) using data sets that combined all species observed over all sampling dates. The Routledge (1977) index ( $\beta_R$ ) was calculated for all six pairwise site combinations and as an overall (all 4 sites) indicator of compositional heterogeneity. This index ranges from 0 to 1, where higher values represent greater community differences among wetland types. The classic index of beta diversity is Whittaker's  $\beta_w$  (Whittaker 1960). McCune and Grace (2002) suggested that  $\beta_w < 1$  and  $\beta_w > 5$  be interpreted as low and high beta diversity, respectively. Overall metrics were determined with and without inclusion of the moist-soil site to indirectly test whether this site was driving the results, based on expected high species overlap with the immediately adjacent perimeter ditch.

The ditch and moist-soil sites were sampled together on 12 dates (see Odonata census subsection). We used multi-response permutation procedures (MRPP) to supplement the descriptive indices and test the null hypothesis of no significant difference in odonate composition and relative abundance of dragonflies between the ditch and moist-soil marsh during the 12 same-date samples. Relative abundances, rather than absolute counts, were used to minimize effects of collecting data from different sized plots (Table 1). Bray-Curtis distances were calculated from matrices of sample date by both total and shared-only species. The total analysis used the total assemblage of each site, while the shared-only analysis included only those taxa observed at least once in both sites. Our interest in the latter case was the observation patterns for the overlapping assemblages; although some species were seen at both sites, the relative abundance (dragonflies only) structure or timing of encounter may differ between the sites. The MRPPs were performed using PC-Ord Version 4.0 (McCune and Medford 1999).

Results

The mean ( $\pm 1$  s.d.) over the full sample period for air temperature was  $31.3 \pm 2.45$  °C, for wind speed was  $0.72 \pm 0.40$  m s<sup>-1</sup>, and for relative humidity was  $53.17 \pm 9.02\%$ . Modal cloud cover was mostly sunny (5–30%) with an average of partly cloudy (30–60%). Weather data were pooled from the moist-soil and ditch sites (n = 17 observations each) and all variables were similar between these sites (p > 0.1, t-tests).

In the ditch, vegetation was sampled during early June in 37 randomly spaced quadrats along the plot centerline, yielding an average richness of 4.0 spp. m<sup>-2</sup> and aerial cover of 63.3%. The ditch supported a minimum of 33 plant species. Herbaceous strata generally occupied two maximum vertices: 1.75–2.0 m canopy of the grasses *P. scoparium* and *P. urvillei*, and a  $\leq 1.5$  m profile of the remaining herbs. There were dense pockets (> 75% cover) of *Juncus coriaceus* Mackenzie and *J. effusus* L. in flooded sections. The woodland edge was predominantly *A. rubrum*, *Diospyros virginiana* L., *Fraxinus pennsylvanicum* Marsh., *L. styraciflua*, and *Quercus* spp., interspersed with tree seedlings, mixed forbs, *Rubus* shrubs, and stands of *Erianthus giganteus* (Walt.) Beauv.

Moist-soil plots contained 23 plant species and only 11 overlapped with those encountered in the ditch plot. Mean richness (1.2 spp. m<sup>-2</sup>) and aerial cover (21.6%), as measured from 16 quadrats in four moist-soil plots, were also lower than the ditch. Although one goal of moist-soil management is to reduce woody encroachment, managers left partial shrub cover along the margins where our plots were located. As a result, the moist-soil plots contained more woody vegetation than found in the ditch plot.

Thirty-one Anisoptera (dragonflies) and 11 Zygoptera (damselflies) were identified within all habitats (Table 2), which is approximately 90% of the total species documented at Noxubee NWR, to date (L. Bennett and J. Bried, unpublished data). Twelve species were encountered in only one of the four types of wetlands, compared to 17 found in two, and 9 found in three. The only species documented across all wetland types were *Erythemis simplicicollis*, *Ischnura posita*, *Libellula lydia*, and *Pachydiplax longipennis*,

Table 2. Species of adult Odonata (N = 42) documented across four types of freshwater wetlands at Noxubee National Wildlife Refuge, MS. Families/subfamilies are arranged phylogenetically within suborders according to Needham et al. (2000) and Westfall and May (1996). Species in **bold** font were observed in only one wetland type. X = present. BLH = bottomland hardwoods; MSU = moist-soil management units; PD = perimeter ditch; RBR = Ross Branch Reservoir.

|                                             | BLH | MSU | PD | RBR |
|---------------------------------------------|-----|-----|----|-----|
| Anisoptera (dragonflies)                    |     |     |    |     |
| Aeshnidae                                   |     |     |    |     |
| <i>Anax junius</i> (Drury)                  | –   | X   | X  | –   |
| <i>Epiaeschna heroes</i> F.                 | X   | X   | X  | –   |
| <b><i>Gomphaeschna furcillata</i></b> (Say) | –   | –   | X  | –   |

Table 2, continued.

|                                             | BLH  | MSU  | PD   | RBR  |
|---------------------------------------------|------|------|------|------|
| Gomphidae                                   |      |      |      |      |
| <i>Arigomphus maxwelli</i> (Ferguson)       | —    | —    | X    | X    |
| <i>Dromogomphus spinosus</i> (Selys)        | —    | —    | X    | —    |
| <i>Gomphus exilis</i> (Selys)               | —    | X    | X    | X    |
| Cordulegastridae                            |      |      |      |      |
| <i>Cordulegaster obliqua</i> (Say)          | —    | —    | X    | —    |
| Libellulidae                                |      |      |      |      |
| Macromiinae                                 |      |      |      |      |
| <i>Macromia illinoiensis</i> (Walsh)        | X    | —    | X    | —    |
| <i>M. taeniolata</i> (Rambur)               | X    | —    | X    | —    |
| Corduliinae                                 |      |      |      |      |
| <i>Epitheca cynosura</i> (Say)              | —    | X    | X    | X    |
| <i>E. princeps</i> (Hagen)                  | —    | X    | —    | X    |
| <i>Somatochlora linearis</i> (Hagen)        | X    | —    | X    | —    |
| Libellulinae                                |      |      |      |      |
| <i>Celithemis eponina</i> (Drury)           | —    | X    | —    | X    |
| <i>C. fasciata</i> (Kirby)                  | —    | —    | —    | X    |
| <i>Erythemis simplicicollis</i> (Say)       | X    | X    | X    | X    |
| <i>Erythrodiplax miniscula</i> (Rambur)     | —    | X    | —    | —    |
| <i>Ladona deplanata</i> (Rambur)            | —    | X    | —    | X    |
| <i>Libellula cyanea</i> F.                  | —    | —    | X    | —    |
| <i>L. incesta</i> (Hagen)                   | —    | X    | X    | X    |
| <i>L. luctuosa</i> (Burmeister)             | —    | X    | X    | —    |
| <i>L. lydia</i> (Drury)                     | X    | X    | X    | X    |
| <i>L. pulchella</i> (Drury)                 | —    | X    | —    | —    |
| <i>L. semifasciata</i> (Burmeister)         | —    | X    | X    | —    |
| <i>L. vibrans</i> F.                        | X    | X    | X    | —    |
| <i>Pachydiplax longipennis</i> (Burmeister) | X    | X    | X    | X    |
| <i>Pantala flavescens</i> F.                | —    | X    | X    | —    |
| <i>P. hymenaea</i> (Say)                    | —    | X    | X    | —    |
| <i>Perithemis tenera</i> (Say)              | —    | X    | —    | X    |
| <i>Sympetrum ambiguum</i> (Rambur)          | X    | X    | X    | —    |
| <i>Tramea carolina</i> L.                   | —    | X    | X    | X    |
| <i>T. lacerata</i> (Hagen)                  | —    | X    | X    | —    |
| Zygoptera (damselflies)                     |      |      |      |      |
| Calopterygidae                              |      |      |      |      |
| <i>Calopteryx maculata</i> (Beauvois)       | X    | —    | —    | —    |
| Lestidae                                    |      |      |      |      |
| <i>Lestes disjunctus</i> (Selys)            | —    | X    | X    | —    |
| <i>L. rectangularis</i> (Say)               | —    | X    | —    | —    |
| <i>L. vigilax</i> (Hagen)                   | X    | —    | —    | —    |
| Coenagrionidae                              |      |      |      |      |
| <i>Argia apicalis</i> (Say)                 | X    | —    | X    | X    |
| <i>A. tibialis</i> (Rambur)                 | X    | —    | —    | —    |
| <i>Enallagma signatum</i> (Hagen)           | —    | X    | —    | X    |
| <i>E. traviatum</i> (Selys)                 | —    | X    | —    | —    |
| <i>Ischnura hastata</i> (Say)               | —    | X    | X    | X    |
| <i>I. posita</i> (Hagen)                    | X    | X    | X    | X    |
| <i>I. ramburii</i> (Selys)                  | —    | X    | —    | X    |
| Species richness ( <i>R</i> )               | 14   | 29   | 28   | 19   |
| No. unique species ( <i>U</i> )             | 3    | 4    | 4    | 1    |
| Sample unit uniqueness ( <i>U/R</i> )       | 0.21 | 0.14 | 0.14 | 0.05 |
| Total sample uniqueness ( <i>U/42</i> )     | 0.07 | 0.10 | 0.10 | 0.02 |
| Total sample richness ( <i>R/42</i> )       | 0.33 | 0.69 | 0.67 | 0.45 |

which are among the most commonly observed and widely distributed species in the United States (Donnelly 2004a,b).

The moist-soil and ditch habitats supported almost twice the total species of the reservoir and bottomland combined (Table 2). Compositions among most site pairs were less than 50% similar despite low beta diversity (e.g.,  $\beta_w < 1$ ) for the total floodplain mosaic, calculated with and without inclusion of the moist-soil site data (Table 3). Note that  $\beta_R$  was only slightly greater when moist-soil data were omitted, and  $\beta_w$  did not change.

The perimeter ditch and moist-soil marsh shared 19 of the 38 species found at these sites (Tables 2 and 3). There were significant differences in species occupancy patterns, but relative abundance structure of dragonflies was similar between sites (Table 4).

Table 3. Diversity index scores for all pairwise comparisons of habitats. Site abbreviations are defined in Table 2.

| Habitat comparison         | Index <sup>A</sup> |       |           |           |
|----------------------------|--------------------|-------|-----------|-----------|
|                            | $C_{jk}$           | $C_s$ | $\beta_R$ | $\beta_w$ |
| BLH vs MSU                 | 0.80               | 0.33  | 0.29      | na        |
| BLH vs PD                  | 0.68               | 0.49  | 0.12      | na        |
| BLH vs RBR                 | 0.82               | 0.31  | 0.26      | na        |
| MSU vs PD                  | 0.50               | 0.67  | 0.19      | na        |
| MSU vs RBR                 | 0.55               | 0.63  | 0.05      | na        |
| PD vs RBR                  | 0.69               | 0.47  | 0.17      | na        |
| Overall $\beta$ -diversity | na                 | na    | 0.26      | 0.87      |
| MSUs excluded <sup>B</sup> | na                 | na    | 0.36      | 0.87      |

<sup>A</sup>Dissimilarity index ( $C_{jk}$ ); Sørensen’s similarity ( $C_s$ ); Routledge’s beta diversity ( $\beta_R$ ); Whittaker’s beta diversity ( $\beta_w$ )

<sup>B</sup>“MSUs excluded” refers to calculations involving BLH, PD, and RBR without MSU

Table 4. Relative abundance of dragonflies and presence-absence patterns of Odonata (includes damselflies) compared between the ditch and moist-soil marsh using multi-response permutation procedures (MRPP). Mean Bray-Curtis distance values are averages for all pairwise combinations of sample dates (12 dates). “All species” is the full species compliment observed in each site and “shared species” were seen in each site at least once, but not necessarily on the same day.

|                     | Mean Bray-Curtis distance |            |       |         |
|---------------------|---------------------------|------------|-------|---------|
|                     | Ditch                     | Moist-soil | $A^A$ | $p$     |
| Dragonfly abundance |                           |            |       |         |
| All species         | 0.379                     | 0.570      | 0.051 | 0.053   |
| Shared species      | 0.412                     | 0.562      | 0.026 | 0.143   |
| Observed odonates   |                           |            |       |         |
| All species         | 0.394                     | 0.395      | 0.208 | < 0.001 |
| Shared species      | 0.467                     | 0.334      | 0.187 | < 0.001 |

<sup>A</sup>Chance-corrected within-group agreement ( McCune and Grace 2002).

## Discussion

### Odonate diversity among mosaic wetlands

Our data show differences of adult odonate communities in several wetland habitats in a freshwater riparian system. For example, 24 species were found in the ditch, including 4 unique species. The ditch vegetation (mean 4.0 spp. m<sup>-2</sup> and 63% aerial cover) likely provided suitable roosting physiognomy away from breeding habitat. In similar studies, a floodplain ditch in England supplied approximately 90% of the odonate richness among four localized riparian habitats (Armitage et al. 2003), and flooded ditches in close proximity at a nature reserve helped maintain invertebrates of high conservation value, including dragonflies (Painter 1999).

The nested man-made wetlands may provide important supplemental habitat resources in the natural floodplain. More species were recorded in each of these wetlands than in the bottomland forest and with < 50% similarity for each pairwise combination involving the bottomland. Low species richness and high uniqueness (Table 2) in the forested wetland area was expected because adult Odonata are ectotherms and prefer sunny areas. Furthermore, forested habitat may obstruct mobility while herbaceous-dominated areas are relatively permeable for adult odonates. It has been shown, for example, that damselflies can travel more quickly across open landscapes than forested areas (Jonsen and Taylor 2000a, Pither and Taylor 1998, Purse et al. 2003). The higher light levels and lack of structural barriers to movement in the ditch, moist-soil marsh, and lake margins likely combine to create favorable habitats unavailable in tracts of surrounding riparian forest. However, without immature stages or final exuviae we cannot confirm that observed species bred in these places, and so they should not be viewed as true resident populations.

### Wetland habitat preferences

Localized maintenance of wetland biodiversity is a function of meso-scale (landscape scale) spatial density, configuration, and composition of ecosystems (Gibbs 1993). The odonate species assemblage at Noxubee NWR appeared to segregate among local floodplain habitats, because 29 of 42 species were observed in only one or two sites and 12 species were seen in only a single wetland type. Worthen (2003) reported site affinity in most of the 41 adult odonate species documented during biweekly visits to six localized bottomland habitats in Congaree Swamp National Park, SC. Kinvig and Samways (2000) found only 14% species overlap among four localized and morphologically distinct (i.e., unique size and vegetative physiognomy) stream sites in South Africa. These results are surprising given the capacity for periodic emigrations, local dispersion, and even large-scale migration (e.g., Russell et al. 1998) in some adult dragonflies. However, movement between habitats of > 1 km is rare in most adult odonates (Conrad et al. 1999, Purse et al. 2003, Stettmer 1996; but see Michiels and Dhondt 1991), especially Zygoptera. Spacing among sites in the current

study was within a 1.5-km radius, although some of the farthest natural bottomlands were > 1.5 km from the man-made habitats.

The segregation of 19 species between the adjacent ditch and marsh sites, in combination with low species overlap (< 50% similarity) among most pairwise comparisons of sites, suggests habitat specialization by adult Odonata. But again, without proof of residence (i.e., larvae or exuviae), we must assume a non-resident distribution. Little or no change in non-gradient metrics ( $\beta_R$ ,  $\beta_W$ ) following removal of the moist-soil site further suggests odonate structural independence among sites. These findings were supported by MRPP analysis of total and shared Odonata in the ditch and moist-soil plots. Compositional asymmetry for the total assemblage is probably linked to the low species overlap because seasonality in adult flight periods varies among species (Wissinger 1988a,b).

Prior to drawdown, species in moist-soil units made oviposition attempts, copulation wheels, and tandem flights, all indicative of reproductively suitable habitat, while ditch species were seen only to roost and forage. The ditch was not hydrologically suitable for breeding activity and thus could not support resident populations. However, it apparently attracted a unique set of transient species that may have been making periodic emigrations (Corbet 1999) from breeding areas.

In addition to hydrology, vegetation is another major mechanism determining the formation of odonate species assemblages (Chwala and Waringer 1996, Schindler et al. 2003, Steytler and Samways 1995) and provides perch sites, territorial boundary markers, oviposition substrate, and emergence platforms for adults (Buchwald 1992). The ditch plot had higher plant richness and aerial cover than the moist-soil plots, and the ditch community was herbaceous while moist-soil plots were mixed herb-shrub assemblages. Wetland sites in close proximity to each other but with varying quantities of vegetation may have markedly different odonate communities (Moore 1991). Dissimilar physiognomic structure and vegetation quantity may have added to differences in species among the sites.

Different sized survey areas or habitat areas (i.e., larger moist-soil habitat area) also may have contributed to compositional differences between sites (Kadoya et al. 2004), although complex requirements of adult odonates can override species-area correlations (J. Bried, unpubl. data; Osborn and Samways 1996).

Observation patterns for shared species may reflect temporal preferences for occupying breeding versus non-breeding habitats. For example, *Celithemis eponina* is reproductively active during mid-morning (Miller 1982) and must therefore occupy breeding habitat at that time. Our observations were limited to 10:00–12:45 hr in the ditch and moist-soil plots. Therefore, natural variation in occupancy, both between and within days and among species, individuals, and within individuals (e.g., between sex, age, or size classes), may have separated “shared” species among these sites.

Relative abundance of dragonflies was consistent between sites over the given gradient. This conflicted with results for presence-absence patterns of overall odonates and suggests that damselflies may have influenced those patterns. However, reanalysis using binary data of only dragonflies still resulted in significant differences between sites (MRPP,  $p < 0.001$ ). It is unclear why relative abundance structure failed to correspond with occupancy patterns.

In recognition of the wide range within a given odonate assemblage, there is need for community-level and autecological studies of habitat associations in wetlands of North America. Future studies of adult odonate distribution should be designed to separate the habitat generalists ("eurytopic") from the specialists ("stenotopic") and shed light on the understudied habitat selection continuum (Corbet 1999, chapter 2 and references therein) in wetland ecosystems. Such studies not only uncover mechanisms behind patterns of organism distribution in nature (MacArthur 1972), but also are potentially useful for detecting practical ecological indicators of ecosystem integrity.

### Acknowledgments

Personnel at NNWR were extremely accommodative. Thanks to Dr. Lloyd Bennett for sharing his odonate faunistic information from NNWR. Dr. Ronald Altig, Dr. Ferenc de Szalay, and anonymous reviewers provided helpful critique of an earlier draft of the manuscript. This research was supported in part by a USGS Water Resources Research grant #01HQGR0088.

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